



NITI Aayog

Strategy for

New India @ 75

STRATEGY FOR NEW INDIA @ 75

MODULE - 3

3

TECHNOLOGY AND INNOVATION



OBJECTIVES



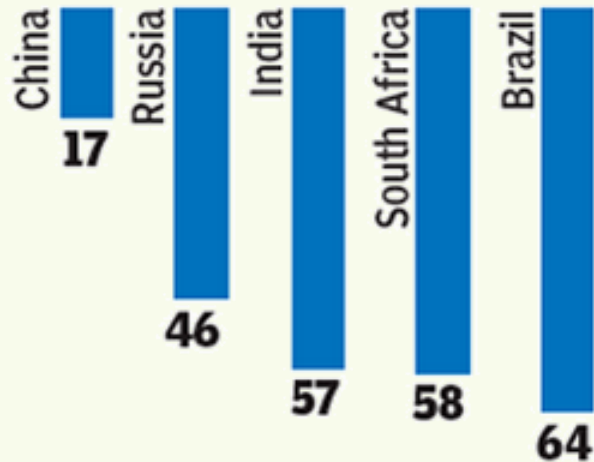
- India should be among the top 50 countries in the Global Innovation Index by 2022-23.
- Five of our scientific research institutions should be among the top 100 in the World.
- India should aim to spend at least 2% of GDP on R&D with equal contributions from the public and private sectors.



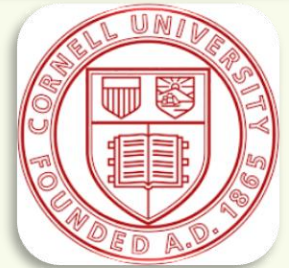
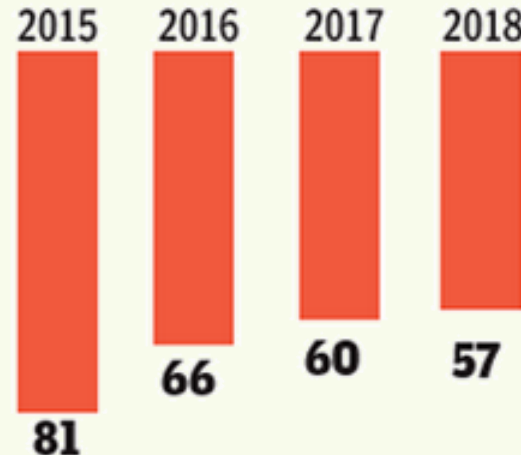
TOP 10 COUNTRIES (Rank in 2018)

- 1 Switzerland 2 Netherlands 3 Sweden 4 UK 5 Singapore
6 USA 7 Finland 8 Denmark 9 Germany 10 Ireland

Ranks of BRICS nations in 2018



India climbs up GII rankings



GII aims to capture the multi-dimensional facets of innovation and provide the tools that can assist in tailoring policies to promote long-term output growth, improved productivity, and job growth.

नीति आयोग
National Institution for Transforming India**1971**

Department of Science and Technology established.
Promotes basic research via:

- Research funding schemes through Science and Engineering Research Board (SERB) and other in-house programmes
- Autonomous Institutions (AIs) under DST were set up

2003

Science and Technology Policy brought together Science and Technology

2016

Atal Innovation Mission launched

2000

National Innovation Foundation was set up to fund grassroots innovations

2013

Science, Technology and Innovation (STI) Policy formulated.



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TECHNOLOGY AND INNOVATION

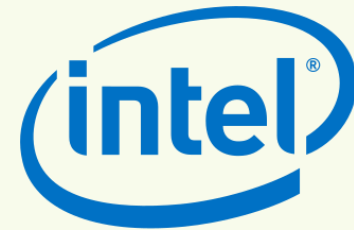
PRESENT SCENARIO



- India became a major destination for outsourced R&D activities.
- India has more than 1,100 R&D centres set up by MNCs, like IBM, Google, Microsoft, Intel etc.
- 37 Indian Scientists from 9 Indian Institutions played a key role in the discovery of gravitational waves that received the Physics Nobel Prize in 2017.
- India is the third largest country in terms of number of start-ups.
- Government set up Atal Innovation Mission to transform radically the innovation, entrepreneurship and start-up ecosystem.

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TECHNOLOGY AND INNOVATION



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TECHNOLOGY AND INNOVATION

CONSTRAINTS

5 The number of scientific R&D professionals in India is 218 per million population and it is abysmally low compared to China's 1,113 and USA's 4,019.

4 The share of the private sector in R&D investment in most technologically advanced countries is as high as 65% to 75%, but, it is only about 30% in India.

**POOR R&D EXPENDITURE**

1 Though R&D expenditure tripled in the period from 2004-05 to 2014-15, its size as percentage of GDP remained at 0.7%.

2 This is very low compared to 2% and 1.2% spent by China and Brazil respectively.

3 Israel spent 4.3% of their GDP on R&D.

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TECHNOLOGY AND INNOVATION

CONSTRAINTS

6 The public procurement system is heavily biased in favour of experienced and established products and technologies.

5 The adoption of indigenous innovations by Indian industry is not very encouraging.

4 But, the rate of transfer of these technologies to industry and for societal benefits is very low.



POOR LINKAGE
BETWEEN HIGHER
EDUCATION &
INDUSTRY



1 The link between research, higher education and industry is weak and nascent.

2 Education system not focused on cultivating scientific temperament at an early age.

3 Renowned public funded institutions like CSIR, DRDO, BARC, ISRO developed frontline technologies.

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TECHNOLOGY AND INNOVATION

WAY FORWARD

EMPOWERED BODY FOR CO-ORDINATION IS NEEDED!



- An empowered body is needed to steer holistically the management of science in the Country.
- It should look at inter-ministerial and inter-disciplinary research.
- There is major weakness of publicly funded R&D institutions like CSIR, ISRO, BARC, DRDO for their poor marketing skills and information dissemination.

3 TECHNOLOGY AND INNOVATION

WAY FORWARD

HOW TO ENHANCE TECHNOLOGY COMMERCIALIZATION OF PUBLIC FUNDED INSTITUTIONS?



- 1 Value addition centres may be set up to upscale the technologies and improving technologies with demonstration of industrial scale pilot production.
- 2 Coordinating with investors to incubate entrepreneurs.
- 3 Bridging the gap between industry and technology development teams.

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TECHNOLOGY AND INNOVATION

WAY FORWARD

HOW TO ENHANCE TECHNOLOGY COMMERCIALIZATION
OF PUBLIC FUNDED INSTITUTIONS?

4

DST should create a National Technology Data Bank in coordination with all public funded R&D institutions. This will provide a central database for technologies that are ready for deployment.

5

Public funded research institutions should consider shifting their focus to the development and deployment of socially relevant technologies in areas like clean drinking water, sanitation, energy, organic farming etc.

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TECHNOLOGY AND INNOVATION

WAY FORWARD



- Frugal innovation provides functional solutions through few resources for the many, who have little means.
- It is the practice generally in emerging markets.
- It purposefully targets the bottom and then makes its way up to other levels to benefit all users.

OTHER SUGGESTIONS

- In Government procurements, preference should be given to promote Make in India initiative.
- A non-lapsable “District Innovation Fund” with a corpus of about Rs 2 Cr in each district may be created to promote grassroot innovations.
- Atal Innovation Mission has already launched Atal New India Challenges in partnership with five ministries in areas of national importance such as solid waste management, water and wastewater management as well as road and rail transport.
- Atal Incubation Centres will provide the platform for promoting frugal innovation.

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TECHNOLOGY AND INNOVATION

WAY FORWARD

OTHER SUGGESTIONS ... *Contd*

- Atal Innovation Mission has set up over 1,000 Atal Tinkering Labs around the country covering over 625 districts.
- It aims to take this number to 10,000 by 2020.
- Further expansion will be considered based on the outcome of the first phase.
- To promote entrepreneurship and start-ups, AIM is supporting Atal Incubation Centres across the Country.
- It is expected that more than 100 World-class Incubation Centres will be up and running by 2020.

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TECHNOLOGY AND INNOVATION

WAY FORWARD

OTHER SUGGESTIONS

... *Contd*

Genomics, study of the structure, function, and inheritance of the genome (entire set of genetic material) of an organism.

- Foreign collaborations, consultants, visiting faculty need to be involved in pursuing R&D in the emerging areas of basic sciences, such as nano-technology, stem cell research, astronomy, genetics, next generation genomics, drug discovery etc.
- Higher Education Commission may consider giving credits for innovation and start-ups and also setting up online entrepreneurial development courses in colleges and universities.

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SINCERE THANKS
FOR VIEWING THIS VIDEO

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